STARD 2015 reporting checklist

Manuscript: "Bias toward the assisting algorithm in polysomnography reference standards built from edited automatic pre-scoring". Page numbers refer to the main document as built on 10 November 2026 (page 1 is the title page).

How STARD applies. This is a methodological study of reference standards, not a classical diagnostic accuracy study. The index tests are automatic scoring algorithms: the pre-fill that the experts edited, and two independent stagers that no expert saw (YASA and GSSC). The target is the label of each epoch or scored event. Accuracy is agreement (Cohen's kappa for stages, event-level F1 for events) with two expert consensus references: one made without the pre-fill (manual) and one made by editing it (assisted). Items about patients' clinical characteristics are reported as far as the open dataset documents them.

Section	No.	STARD 2015 item	Where reported
Title or abstract	1	Identification as a study of diagnostic accuracy using at least one measure of accuracy (such as sensitivity, specificity, predictive values, or AUC)	Title (p. 1) and abstract (p. 3): a study of the reference standards used to validate automatic scoring, with the paired design (the same experts scoring manually and by editing the pre-scoring) in the abstract; the accuracy measures, Cohen's kappa and event-level F1, are in Methods: Agreement and reference standards (p. 5)
Abstract	2	Structured summary of study design, methods, results, and conclusions	Abstract (p. 3). JSR requires unstructured abstracts; the single paragraph covers design, methods, results and conclusions
Introduction	3	Scientific and clinical background, including the intended use and clinical role of the index test	Introduction (pp. 3-4)
	4	Study objectives and hypotheses	Introduction, last paragraph (p. 4); hypotheses and predictions in Supplementary File S1 and Supplementary Table S10
Methods: study design	5	Whether data collection was planned before the index test and reference standard were performed (prospective study) or after (retrospective study)	Methods: Data, and Analysis plan and estimand (pp. 4-5). Secondary analysis of an existing open dataset, under analysis plans fixed before computing
Methods: participants	6	Eligibility criteria	Methods: Data (p. 4). The 20 recordings chosen by the dataset's creators; all were analysed
	7	On what basis potentially eligible participants were identified (such as symptoms, results from previous tests, inclusion in registry)	Methods: Data (p. 4). Clinical polysomnograms selected by the dataset's creators at five percentiles of automatic-versus-clinical agreement (the 12.5th, 37.5th, 50th, 62.5th and 87.5th), all accepted for manual scoring in the routine workflow
	8	Where and when potentially eligible participants were identified (setting, location and dates)	Methods: Data (p. 4). Haaglanden Medisch Centrum Sleep Center, The Hague, the Netherlands; recorded in 2019
	9	Whether participants formed a consecutive, random or convenience series	Methods: Data (p. 4). Selected by the dataset's creators by the percentile rule above; not consecutive or random
Methods: test methods	10a	Index test, in sufficient detail to allow replication	Methods: Data (pre-fill algorithms, p. 3) and Secondary analyses (YASA 0.7.0 and GSSC 0.0.9, p. 5); Supplementary Methods (algorithm references, stager channels and settings)
	10b	Reference standard, in sufficient detail to allow replication	Methods: Data (12 technicians, AASM manual 2.4, WASM 2016, scoring sessions, p. 4) and Agreement and reference standards (consensus rules, p. 5); Figure 1 (p. 5)
	11	Rationale for choosing the reference standard (if alternatives exist)	Introduction, first paragraph (p. 3); Methods: Analysis plan and estimand (p. 5), which states the estimand; Discussion: What the data establish (p. 13)

Section	No.	STARD 2015 item	Where reported
	12a	Definition of and rationale for test positivity cut-offs or result categories of the index test, distinguishing pre-specified from exploratory	Methods: Agreement and reference standards (p. 5): five sleep stages (W, N1, N2, N3, R) and scored event intervals, as in the AASM and WASM rules; pre-specified (Supplementary File S1)
	12b	Definition of and rationale for test positivity cut-offs or result categories of the reference standard, distinguishing pre-specified from exploratory	Methods: Agreement and reference standards (p. 5): at least 7 of 12 experts (6 of 11 leave-one-out); robustness variants in Supplementary Methods; all pre-specified
	13a	Whether clinical information and reference standard results were available to the performers/readers of the index test	Methods: Data and Secondary analyses (pp. 3, 5). The index tests are algorithms; the stagers had no access to any scoring, and the pre-fill was produced before the assisted session
	13b	Whether clinical information and index test results were available to the assessors of the reference standard	The subject of the study. Methods: Data (p. 4) and Figure 1 (p. 5): the manual reference was scored blinded to earlier scorings and without the pre-fill; the assisted reference was made by editing the pre-fill; no expert saw either stager. Discussion: Recommendations (p. 14)
Methods: analysis	14	Methods for estimating or comparing measures of diagnostic accuracy	Methods: Agreement and reference standards, Primary measures and inference, Secondary analyses, Unassisted audit (pp. 5-7)
	15	How indeterminate index test or reference standard results were handled	Methods: Agreement and reference standards (p. 5): unscored epochs dropped from each comparison; epochs without a consensus have no reference label; common-support analyses (Secondary analyses, p. 5)
	16	How missing data on the index test and reference standard were handled	Supplementary Methods: one recording without the C4-M1 channel used Cz-M1 for both stagers; rules for empty event scorings
	17	Any analyses of variability in diagnostic accuracy, distinguishing pre-specified from exploratory	Methods: Secondary analyses (pp. 6-7): by task, stage, event type, manual agreement and expert; robustness variants; negative control. Supplementary Table S10 lists all predictions and outcomes; the one configuration examined after the fact is labelled (Supplementary Methods)
	18	Intended sample size and how it was determined	Methods: Primary measures and inference (p. 6). Fixed by the open dataset (20 recordings, 12 experts); no sample size calculation; conclusions limited to these experts and recordings
Results: participants	19	Flow of participants, using a diagram	Figure 1 (p. 5) shows the scorings of each recording. All 20 recordings and all 24 scorings of each were analysed; none was excluded
	20	Baseline demographic and clinical characteristics of participants	Methods: Data (p. 4): 20 patients, 12 men, median age 57 years (interquartile range 51.8-61.5), median time in bed 7.3 hours, as reported by the dataset's creators; no clinical data per recording are available
	21a	Distribution of severity of disease in those with the target condition	Not applicable: the targets are epoch and event labels, not a diagnosis. Clinical indices (events per hour of sleep) and respiratory severity categories are compared between references (Results: Clinical indices, p. 9; Supplementary Table S12)
	21b	Distribution of alternative diagnoses in those without the target condition	Not applicable, as for 21a
	22	Time interval and any clinical interventions between index test and reference standard	Methods: Data (p. 4): the assisted session followed the manual session by at least four months; no interventions (recordings were scored, not repeated). Discussion: Limitations (p. 14)

Section	No.	STARD 2015 item	Where reported
Results: test results	23	Cross tabulation of the index test results (or their distribution) by the results of the reference standard	Results (pp. 8-12): agreement of each index test with each reference by recording and task (Tables 1-3; Supplementary Tables S2-S4 and S11-S14); side-taking counts (Figure 3c)
	24	Estimates of diagnostic accuracy and their precision (such as 95% confidence intervals)	Results (pp. 8-12), Tables 1-3 and Figures 2-5, with 95% crossed-bootstrap intervals or exact standard errors
	25	Any adverse events from performing the index test or the reference standard	Not applicable: secondary analysis of existing recordings
Discussion	26	Study limitations, including sources of potential bias, statistical uncertainty, and generalisability	Discussion: Limitations (p. 14)
	27	Implications for practice, including the intended use and clinical role of the index test	Discussion: Consequences for validation and Recommendations (pp. 13-14); Conclusions (p. 15)
Other information	28	Registration number and name of registry	Not registered. Methods: Analysis plan and estimand (p. 5); the dated plans are reproduced in Supplementary File S1
	29	Where the full study protocol can be accessed	Supplementary File S1 (protocol and four amendments, verbatim, with times and commit identifiers)
	30	Sources of funding and other support; role of funders	Title page: funding statement (no specific funding)

Checklist items from STARD 2015 (Bossuyt et al., BMJ 2015;351:h5527), which is distributed for free use.